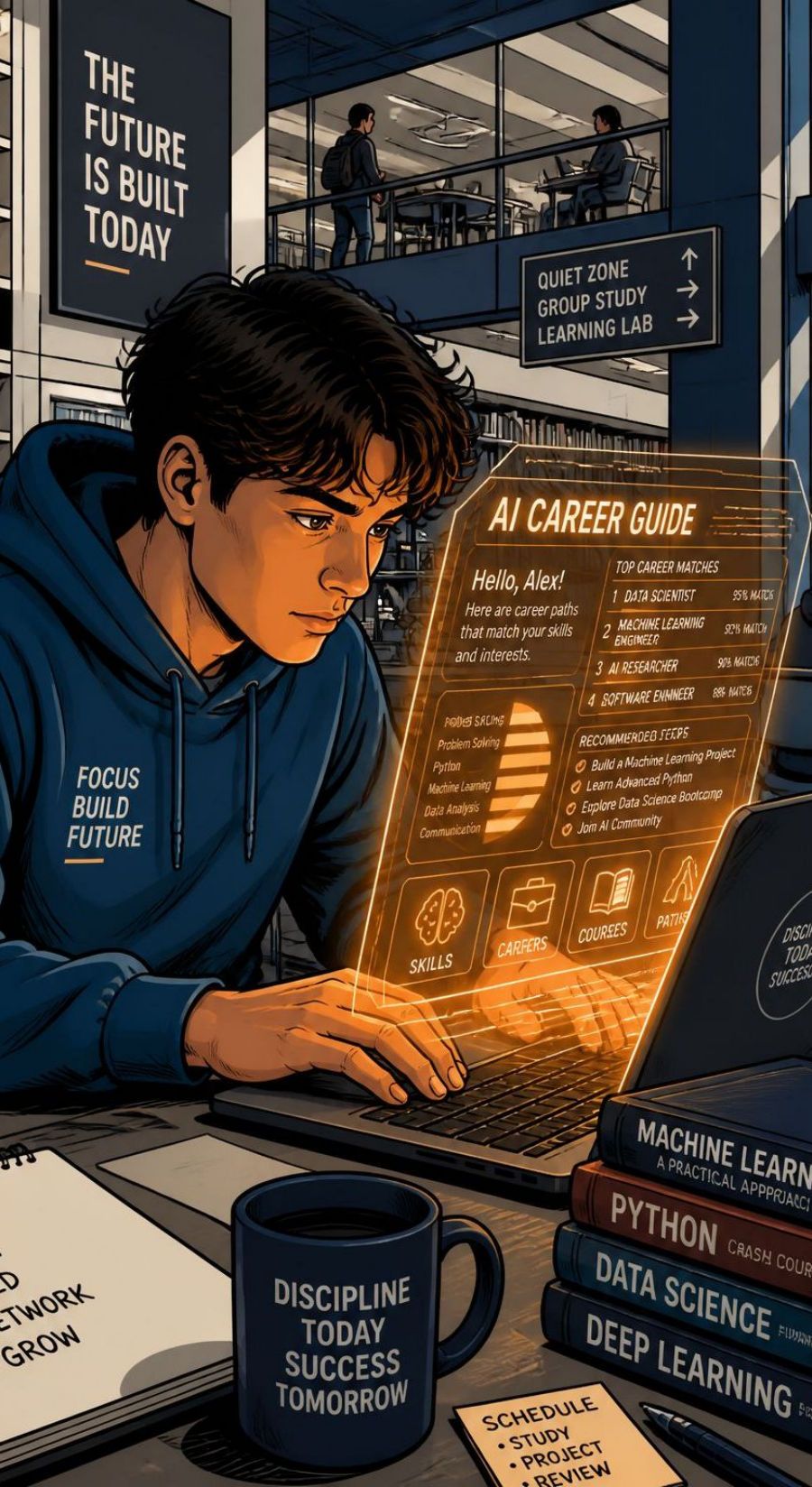




# CareerAI: AI-Powered Career Guidance & Student Success Platform

Empowering students with intelligent, personalized career pathways through AI-driven recommendations and automated workflows.

*Prepared by: Simran Kumari*





# The Problem Students Face

## No Clear Direction

Students struggle to identify which career paths align with their strengths, interests, and market realities.

## Limited Awareness

Awareness of emerging roles, required skills, and industry opportunities remains critically low.

## No Personalized Guidance

Generic advice fails to address individual skill gaps, learning pace, and career aspirations.

## Manual, Slow Processes

Career counseling is resource-intensive and cannot scale to meet growing student demand.

# The CareerAI Effect: From Confusion to Clarity

No more career crossroads – just clear directions ahead!

# What CareerAI Sets Out to Achieve

Six core objectives drive the platform's design — from intelligent recommendations to counselor collaboration.

1

## Personalized Career Recommendations

AI-matched roles based on student profiles and assessments.

2

## Skill Gap Analysis

Identify missing competencies for target careers.

3

## Learning Roadmap Generation

Step-by-step upskilling plans tailored to each student.

4

## Internship & Job Recommendations

Curated opportunities aligned to skills and interests.

5

## Progress Tracking

Monitor milestones and skill development over time.

6

## Counselor Support

Automated reports to enable informed human guidance.





## PROPOSED SOLUTION

# Introducing CareerAI

CareerAI is an end-to-end intelligent platform that combines the power of Google Gemini AI with n8n workflow automation to deliver a seamless, personalized career guidance experience.

→ **AI-Powered Guidance**  
Google Gemini analyzes student profiles and generates tailored career recommendations.

→ **Workflow Automation**  
n8n orchestrates data flow from registration to report delivery — zero manual intervention.

→ **Automated Reports**  
Personalized PDF reports and email summaries delivered directly to students and counselors.

# Technology Stack

CareerAI is built on a robust, cloud-native stack designed for scalability and seamless integration.



## n8n

Core workflow orchestration engine connecting all services.



## Google Gemini API

Generative AI for career analysis and personalized recommendations.



## Google Forms & Sheets

Student data capture and structured storage.



## Gmail

Automated delivery of reports and notifications.



## Database

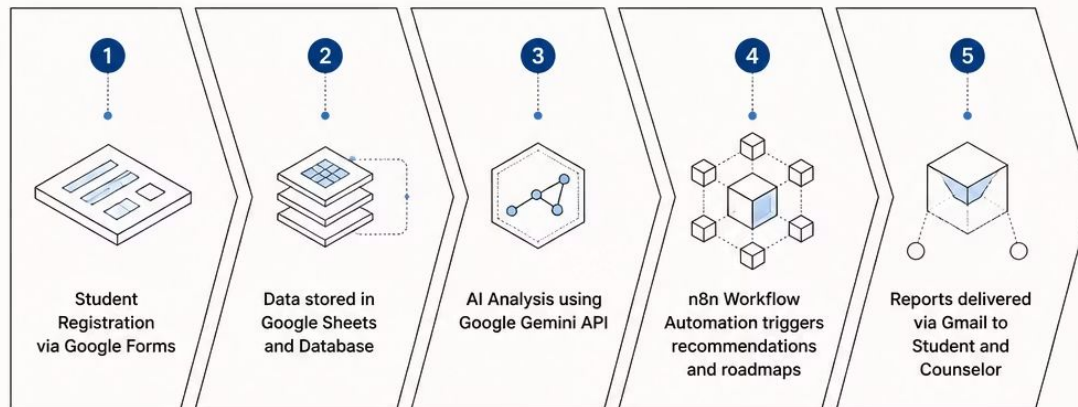
Persistent storage for student records and progress data.



## GitHub

Version control and collaborative development.

# System Architecture



## How Data Flows

The pipeline begins when a student submits the Google Form. n8n captures the response, stores it in Google Sheets, and triggers the AI recommendation engine. Google Gemini processes the profile, generates a career report and learning roadmap, and dispatches everything via Gmail — all within seconds.

□ All five workflows operate independently but share the same data backbone, ensuring consistency and traceability.

# The Five n8n Workflows

01

## Student Registration & Career Assessment

Captures student responses via Google Forms and stores data in Sheets.

02

## AI Career Recommendation Engine

Google Gemini analyzes profiles and returns personalized career matches.

03

## Learning Roadmap Generation

AI creates a step-by-step upskilling plan tailored to each student's target role.

04

## Internship & Job Recommendation

Curates relevant opportunities based on skills, location, and experience level.

05

## Counselor Review & Progress Report

Generates and emails structured progress reports to counselors and students.



# Features: Implemented & Planned

## ✓ Implemented

### AI-Powered Decision Making

Google Gemini drives all career and roadmap recommendations with contextual intelligence.

### Conditional Branching

n8n routes data dynamically based on student responses and assessment outcomes.

## Future Enhancements

- **Human Approval Gates** — Counselor sign-off before final report delivery
- **Error Handling & Logging** — Robust failure recovery and audit trails
- **Cron Scheduling** — Automated periodic progress check-ins
- **Webhook Integration** — Real-time triggers from external systems

# Testing & Results

All five workflows were tested end-to-end with real student data. Every stage executed successfully, producing actionable outputs at each step.

5

Workflows Tested

All executed without errors across test scenarios.

100%

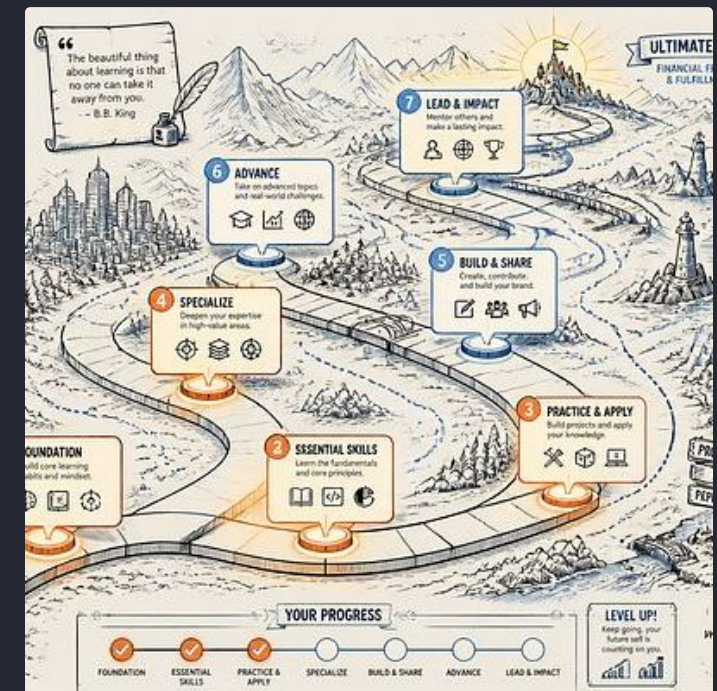
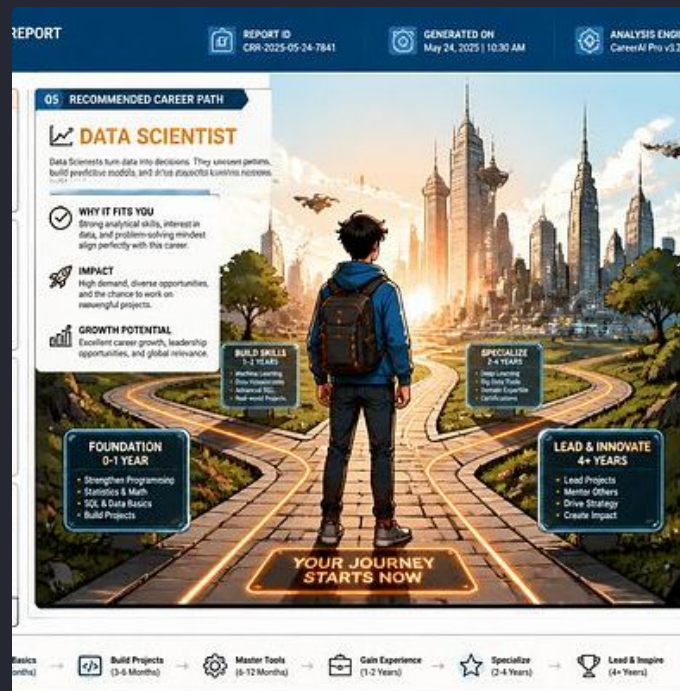
AI Recommendations

Career matches generated for every test profile.

5

Outputs Delivered

Roadmaps, internships, and reports auto-generated and emailed.



# Challenges, Future Scope & Conclusion

## ⚡ Challenges Faced

### AI + n8n Integration

Configuring Gemini API calls within n8n required careful prompt design.

### Workflow Logic Design

Mapping complex career decision trees into conditional branches.

### Prompt Engineering

Iterating on prompts to produce structured, actionable AI outputs.

## 🚀 Future Scope

- LinkedIn API integration for real-time job matching
- Live internship APIs for dynamic opportunity feeds
- Mobile application for on-the-go student access
- Analytics dashboard for counselors and admins
- Scheduled automated progress reports

✅ **Conclusion:** CareerAI demonstrates that AI-powered automation can transform career guidance — delivering personalized, scalable, and actionable recommendations that help students make informed decisions with confidence.